ORSK, Russian Federation

WMO: **351380**

Lat: **51.067N** Lon: **58.600E** Elev: **285** StdP: **97.95** Time Zone: **5.00 (E05)** Period: **04-19** WBAN: **99999**

Annual Heating, Humidification, and Ventilation Design Conditions

	Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF	
				99.6%			99%			0.4%		1%					
		99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(1)	1	-28.8	-26.1	-32.1	0.2	-28.2	-29.8	0.2	-26.2	14.3	-4.7	12.8	-5.2	3.4	30	0.692	(1)

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

	Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB		
			0.4%		1%		2%		0.4%		1%		2%				
			DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD	
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(2)	7	12.9	34.7	18.2	32.9	17.9	31.1	17.5	20.2	29.7	19.4	28.9	18.5	28.0	5.5	130	(2)

	Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB	
	0.4%			1%			2%			0.4%		1%		2%			
	DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(3)	17.1	12.6	23.0	16.1	11.8	22.6	15.0	11.0	22.4	59.2	30.0	56.1	28.3	53.4	27.7	23.4	(3)

Extreme Annual Design Conditions

	Extreme Annual WS				Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
					Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
	1%	2.5%	5%		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
	(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4)	12.3	10.8	9.6	DB	-31.1	37.4	3.0	1.6	-33.2	38.5	-35.0	39.4	-36.7	40.3	-38.9	41.4	(4)
(5)				WB	-31.3	22.0	3.0	1.0	-33.4	22.7	-35.1	23.3	-36.8	23.8	-39.0	24.6	(5)

Monthly Climatic Design Conditions

				Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
				(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg		5.1	-15.1	-14.1	-5.1	7.3	15.5	20.5	22.3	21.4	14.4	6.0	-2.8	-10.5	(6)
(7)		DBStd		14.56	6.90	6.52	5.71	5.97	4.70	4.41	3.63	4.43	4.73	5.04	6.37	6.71	(7)
(8)		HDD10.0		3229	779	674	469	119	8	1	0	0	13	144	384	637	(8)
(9)		HDD18.3		5224	1038	908	727	332	114	28	9	22	134	383	634	895	(9)
(10)		CDD10.0		1430	0	0	0	38	179	316	380	352	146	19	0	0	(10)
(11)		CDD18.3		383	0	0	0	1	26	93	131	116	17	0	0	0	(11)
(12)		CDH23.3		4545	0	0	0	23	336	1113	1491	1383	197	2	0	0	(12)
(13)		CDH26.7		1922	0	0	0	2	90	478	644	642	66	0	0	0	(13)
(14)	Wind	WSAvg		4.8	4.8	4.9	5.5	5.3	5.0	4.6	4.5	4.4	4.5	4.9	5.0	4.7	(14)
(15)	Precipitation	PrecAvg		342	25	19	28	28	33	38	43	32	19	26	24	27	(15)
(16)		PrecMax		506	43	40	65	75	73	97	82	89	46	49	52	56	(16)
(17)		PrecMin		209	5	3	1	7	5	13	9	5	4	1	1	7	(17)
(18)		PrecStd		85	10	10	17	17	18	20	25	25	14	13	13	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB		1.8	1.4	12.1	26.2	31.0	35.9	37.0	37.1	32.9	22.8	10.9	3.1	(19)
(20)			MCWB		1.2	0.9	7.1	13.6	16.7	18.3	18.3	18.3	16.1	11.6	7.2	1.2	(20)
(21)		2%	DB		-0.8	0.1	5.9	22.8	29.1	33.8	34.1	34.2	28.1	19.1	8.8	1.1	(21)
(22)			MCWB		-1.3	-0.4	3.1	12.3	15.8	17.9	18.3	18.1	14.3	9.7	6.2	0.4	(22)
(23)		5%	DB		-2.1	-1.8	3.0	19.9	27.2	31.9	32.1	32.8	25.1	16.2	6.8	0.1	(23)
(24)			MCWB		-2.6	-2.3	1.5	11.4	15.4	17.3	18.1	18.0	13.9	9.1	5.1	-0.6	(24)
(25)		10%	DB		-4.8	-4.0	1.8	16.9	25.0	29.3	30.2	30.2	22.9	13.2	4.6	-1.1	(25)
(26)			MCWB		-5.3	-4.4	0.7	9.9	14.7	17.0	17.6	17.3	12.9	8.1	3.3	-1.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB		0.7	1.0	7.2	15.2	19.2	21.2	21.3	21.5	17.1	13.5	8.5	1.9	(27)
(28)			MCDB		1.2	1.3	11.5	23.6	27.5	30.9	29.9	31.8	27.2	18.8	9.6	2.3	(28)
(29)		2%	WB		-1.3	-0.6	3.2	13.3	17.5	19.8	20.1	20.1	15.7	11.3	6.5	0.4	(29)
(30)			MCDB		-0.9	-0.2	5.2	20.9	26.0	29.3	29.1	30.4	24.9	16.4	7.7	0.8	(30)
(31)		5%	WB		-3.0	-2.3	1.6	11.8	16.3	18.7	19.3	18.9	14.6	9.8	4.7	-0.8	(31)
(32)			MCDB		-2.4	-1.8	2.5	18.7	24.7	28.1	28.5	28.8	23.4	14.9	6.2	-0.2	(32)
(33)		10%	WB		-5.3	-4.5	0.5	10.3	15.3	17.8	18.5	18.0	13.6	8.5	3.1	-2.1	(33)
(34)			MCDB		-4.8	-4.0	1.3	16.1	23.4	26.9	27.9	28.4	21.6	12.8	4.4	-1.4	(34)
(35)	Mean Daily Temperature Range		MDBR		7.1	7.7	7.8	11.0	12.9	13.0	12.9	13.2	12.1	9.5	6.4	6.8	(35)
(36)		5% DB	MCDBR		5.8	6.6	8.6	15.1	15.2	15.8	15.1	15.5	14.8	14.2	8.6	6.0	(36)
(37)			MCWBR		5.8	6.4	6.7	7.8	6.6	6.2	5.7	6.0	6.5	7.6	6.4	5.6	(37)
(38)		5% WB	MCDBR		6.2	6.4	7.9	14.1	13.8	13.8	13.4	14.4	13.0	12.2	7.5	5.6	(38)
(39)			MCWBR		6.2	6.3	6.5	7.9	6.8	6.3	6.0	6.3	6.5	7.4	6.1	5.5	(39)
(40)	Clear-Sky Solar Irradiance	taub			0.295	0.324	0.381	0.418	0.406	0.403	0.410	0.425	0.375	0.355	0.331	0.297	(40)
(41)		taud			2.085	2.018	1.964	2.100	2.231	2.302	2.292	2.221	2.320	2.287	2.188	2.089	(41)
(42)		Ebn at Noon			689	771	793	817	848	853	840	803	807	743	644	615	(42)
(43)		Edh at Noon			80	113	146	143	132	124	124	126	102	87	73	69	(43)
(44)	All-Sky Solar Radiation	RadAvg			1.34	2.51	3.87	4.76	5.92	6.53	6.18	5.30	3.83	2.11	0.97	0.86	(44)
(45)		RadStd			0.14	0.14	0.42	0.46	0.51	0.43	0.33	0.39	0.31	0.34	0.11	0.15	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	+0.98	N/A	N/A	N/A	N/A	+105	+84	(47)

Nomenclature: See separate page